Beyond Parameter Count: A Design Comparison of Granite 4.1, Kimi K2, MiniMax-M2, and Nemotron

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ABSTRACT

Modern open-source LLMs are converging toward high competition with proprietary in regard to quality and performance. This report presents a comparison of four open-source LLMs: Granite 4.1, Kimi K2, MiniMax-M2, and Nemotron. The comparison is based on a set of design criteria that are relevant to the deployment and performance of these models in real-world applications. The report provides an overview of the methodology used for the comparison, as well as a detailed analysis of the architecture and deployment characteristics of each model. The findings highlight the differences in design of each model. Each model presented uses a different approach to model design, which has implications for their performance and suitability for different applications. The report concludes with a discussion of the implications of the findings for the development and deployment of open-source LLMs, as well as recommendations for future research in this area.

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1 INTRODUCTION

To begin the comparison of these four open-source LLMs it is important to first discuss why parameter count is no longer enough to evaluate a models architecture and predict its performance. Modern LLMs are converging in performance but diverge in chosen architecture. Many frontier models utilize dense transformer archiectures while others pursue different training approaches entirely, utilizing hybrid or Mixture of Experts (MoE) architectures. As the industry continues to discover the limitation applied by dense transformer architectures, new and creative solutions are being developed to improve training efficiency and inference performance on less hardware.

1.1 WHY PARAMETER COUNT IS NO LONGER ENOUGH

Parameter count has previously been used as a rough estimation of how a model will perform in given tasks. Historically, language models often followed a linear trend in regard to model performance over a large range of tasks and scale alone. Many modern architectures introduce additional design complexities that contribute to generalization and performance capabilities including sparse expert routing, activated parameter count, hybrid sequence modeling, quantization strategies, context-efficiency, and post-training methodologies.

As a result of this increasing complexity, total parameter count can now be a misleading metric mistaken for a model's performance. Modern design considerations now require introductions of tradeoffs in cost, memory usage, computation time, and inference efficiency, causing a difficult comparison in regard to model performance.

This report compares multiple recent model families represented by these four models:

1. IBM - Granite 4.1
2. Moonshot AI - Kimi K2
3. MiniMax - MiniMax-M2
4. NVIDIA - Nemotron-H

1.2 CHOSEN MODELS FOR COMPARISON

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1.2.1 Granite 4.1

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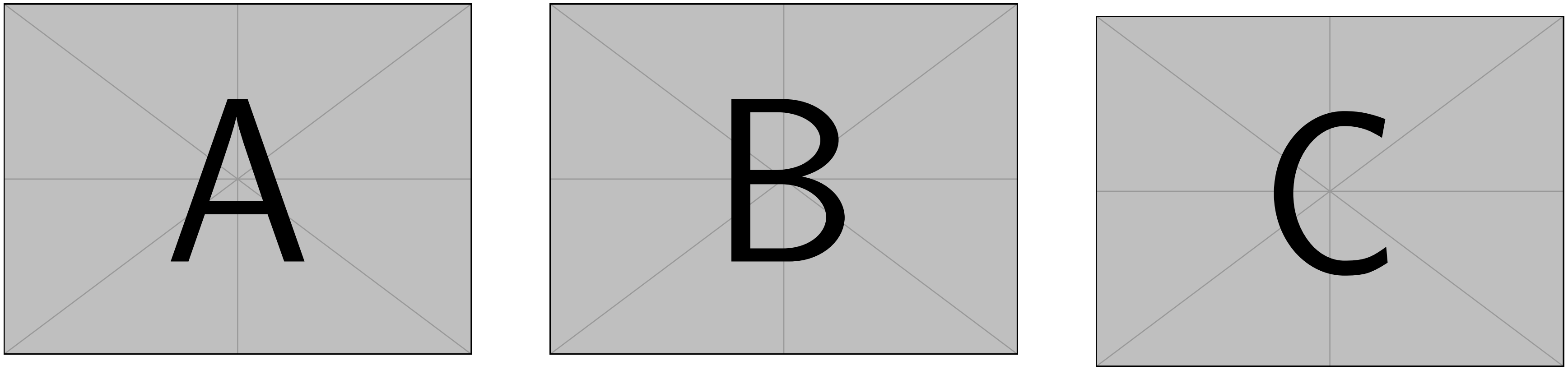

(a) Optional subcaption

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Figure 1.1. A much longer figure description, one that is too long for the table of contents snippet. This example uses subfigures on a landscape layout. (a) You can give an optional subcaption. (b) Omit the subcaption and just use this main caption for text. (right) You can omit the a/b/c labels and use left/middle/right descriptions in this main caption.

1.2.2 Kimi K2

1.2.3 MiniMax-M2

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2 COMPARISON METHODOLOGY

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5 DISCUSSION

6 CONCLUSION

REFERENCES

APPENDIX A

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A.1 FIRST SECTION

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